

Technology #15: MR Flow Quantification Sensor at Catheter Scale

Technology Domain: Magnetic Resonance (MRI/NMR) + Flow Quantification + Miniaturization

Mechanism: Implantable MR flow sensor that quantifies CSF flow rate in shunt catheter using phase-contrast MRI principles. Miniaturized magnet + RF coil integrated into catheter or external patch.

Maturity: ENGINEERING_DEFINITION

Current State: No physical prototype. Computational model and engineering definition exist.

Key Risk: Miniaturization to catheter scale cannot achieve SNR > 10:1 at any B0 field (physically blocked)

Next Experiment: SNR feasibility test (magnet + coil + phantom)

Buyer Receives: MR flow quantification concept + mechanism description; Larmor + phase-contrast + SNR governing equations; Bench test protocols (SNR, flow accuracy, susceptibility, stability, EMC); External precedent catalog (miniaturized NMR, phase-contrast MRI); Critical UNKNOWN disclosure (miniaturization feasibility, SNR at catheter scale, susceptibility artifacts); This engineering dossier with failure analysis + build plan; Honest assessment: miniaturization to catheter scale may be physically blocked (high-risk concept)

Buyer Must Build: Production magnet design (B0, material, miniaturization); Production RF + gradient coil design; Production signal processing + flow calculation; Manufacturing process for catheter-scale MR sensor; Regulatory submission (likely Class III — novel MRI device); Clinical validation (flow accuracy vs standard); Miniaturization feasibility proof (may be blocked); MRI safety assessment + labeling

Transfer Posture: Sponsored Validation

Target Buyer: MRI/sensor company; neuroshunt manufacturer

CONFIDENTIAL. See full engineering dossier for details.